

Reflective Journal

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Coming into this lab as a group still building foundational knowledge in machine learning, the first challenge was simply understanding what each tool was designed to do and why it mattered. Our group chose Google Colab because it is cloud-based, meaning every team member could access the same notebook from any device without installing anything locally. For beginners like most of us on the team, that accessibility removed a barrier that could have slowed progress before we even wrote a single line of code.

Running the code cells myself was where the real learning happened. Seeing the output appear directly below each cell made it easier to connect what the code was doing to what the result actually meant. I did not just observe the process. I executed it, which made the concepts feel more concrete and less abstract.

One of the most insightful parts of this lab was discovering what each Python library is specifically designed to do and how they work together. Pandas allowed us to load the Iris dataset and organize it into a structured table called a DataFrame, making it easy to read and manipulate. NumPy handled the mathematical calculations, such as computing the mean and standard deviation of sepal length, which would have been far more complex to do manually. Matplotlib brought the data to life visually. Creating both a scatter plot and a bar chart helped me see patterns in the data that numbers alone would not have revealed as clearly. Understanding that these three libraries each serve a distinct purpose, yet combine seamlessly in a workflow, was genuinely eye-opening.

The visualizations themselves provided meaningful insight. In the scatter plot, mapped sepal length against sepal width, setosa showed the largest average sepal width among the three

species and formed its own distinct cluster, clearly separated from the others. This indicates that *setosa* can be identified using sepal measurements alone, since its wider sepals set it apart from *versicolor* and *virginica*. Those two species, on the other hand, overlapped more closely with one another, which told me that sepal measurements may not be enough to reliably distinguish them and that additional features like petal length and width could help a model separate classes that look similar on only two dimensions. Seeing this directly in the plot made the idea of feature selection feel much more concrete. The bar chart confirmed that the dataset was perfectly balanced with 50 samples per species, and we now understand that this kind of balance is important because it prevents a model from being biased toward one class during training.

Overall this lab changed how we think about data preparation. Before Module 2, we assumed machine learning was mostly about the algorithms. Now we understand that loading, exploring, and visualizing data with the right tools is just as critical as the model itself.